

## Sec. 5 Experiments

Table 1: Headline cells, tail-500 over 5 seeds, decaying step-size. *full* sees every group every step. The fixed multiplier  $m/K$  diverges whenever observation rates are non-uniform.

| Instance                        | GAVOT         | unif. avg | $m/K$ | full   |
|---------------------------------|---------------|-----------|-------|--------|
| Adult, prevalence ( $\nu=.60$ ) | <b>0.2269</b> | 0.2479    | 0.670 | 0.2073 |
| Adult, aligned ( $\nu=0$ )      | 0.2075        | 0.2077    | div.  | 0.2073 |
| IMDb, skewed ( $\nu=.31$ )      | <b>86.06</b>  | 91.63     | 8504  | 82.95  |
| IMDb, aligned ( $\nu=0$ )       | <b>84.28</b>  | 86.37     | div.  | 82.95  |

### 1 EXPERIMENTS

**Instances.** *Adult* income classification [33]: logistic regression ( $d=109$  with intercept, cross-entropy,  $\eta=0.1$ ) on 100 race-homogeneous groups of 30 samples, the number of groups per race proportional to its prevalence (White/Black/Asian-Pac./Amer-Ind./Other = 85/10/3/1/1); the importance target is the fairness target itself,  $1/5$  per race divided equally among its groups, so a group from the smallest race carries  $p_i = 0.2$ . *IMDb-Wiki* age regression [34]: linear regression of age on 128-d ResNet face embeddings ( $\eta=10^{-2}$ ) over the first 100 identities with 30 images, with the cubic importance target  $p_i \propto (101 - i)^3$ . Observation rates are swept through a one-parameter family: on Adult  $r \propto p^\beta$ ,  $\beta \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$ , from uniform observation (prevalence regime,  $\nu=.60$ : the five groups of the three smallest races hold 60% of the target against  $r_i \approx K/m$ ) to importance-aligned ( $\nu=0$ ); on IMDb-Wiki  $r_i \propto i^\beta$ ,  $\beta \in \{0, \frac{1}{2}, 1, \frac{3}{2}, 2, 3\}$ , from uniform observation ( $\nu=.31$ , nine groups) to the mirror image of  $p$  ( $\nu=.88$ , 41 groups), plus the aligned  $r_i \propto 101 - i$  ( $\nu=0$ ). Eleven instances,  $\nu$  from 0 to .88.

**Protocol.** Each step observes  $K=3$  groups drawn without replacement with probabilities  $\propto r$ , which induces  $q$  over the  $\binom{100}{3}$  compositions;  $q$  is estimated from  $10^6$  draws of the sampler and the plan from (7) on the membership mask, once per instance. Each observed group takes  $H=5$  local steps on its own data before the weighted average, the standard practical deviation from the one-step analysis, applied to every rule alike since all rules consume the same local updates from the same draw  $A^t$ . Rules: **GAVOT**; *group-blind averaging*, the uniform average over the observed groups (Proposition 2 with  $\tilde{p}_i = r_i/K$ ); the *fixed multiplier*  $m/K$ , the rescaling  $(m/K)p_i$  that presumes uniform observation; and *full*, every group every step weighted by  $p$ , the reference  $\Phi(p)$ . Stepsize  $\eta_t = \eta/(1 + t/1000)$  unless stated, 4000 steps, 5 seeds; we report  $F_p$  averaged over the last 500 steps, mean  $\pm$  std over seeds. On both instances  $\arg \min F_w$  is available to machine precision (closed form for least squares, Newton for the logistic loss), so the floors  $\Phi(p), \Phi(\tilde{p}), \Phi(\bar{p})$  of Theorem 8 are exact rather than estimated, and in every instance with  $\nu = 0$  the iteration (7) converges to a row error below  $3 \cdot 10^{-8}$ , certifying the remaining inequalities of (5).

**The least represented group.** Fig. 2 reads the same runs on the group the target exists for. On Adult, trans-

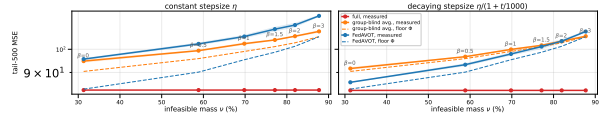


Figure 1: IMDb-Wiki severity sweep. Solid, measured tail-500 ( $\pm 1$  std); dashed, the exact floor  $\Phi(\cdot)$  each rule converges to. Left, constant stepsize: the optimization residual swamps the alignment gain and transport loses everywhere, even where the floors are seven units apart. Right, decaying stepsize: the residual collapses and the measured curves track their floors, so transport wins until the floors close at high  $\nu$ . The sign is set by which of the two terms of (9) dominates, not by the regime label.

Table 2: IMDb-Wiki severity sweep, decaying stepsize.  $\Delta_{\text{floor}}$  is the alignment gain of Theorem 8, *gain* the measured advantage of transport,  $t$  its Welch statistic (population std). The excess residual transport pays averages 1.9, and the sign flips exactly where  $\Delta_{\text{floor}}$  crosses it. The five Adult skews all sit above their crossover and are omitted.

| $\beta$ | $\nu$ | $\Delta_{\text{floor}}$ | gain         | $t$  |
|---------|-------|-------------------------|--------------|------|
| 0       | .31   | 6.99                    | <b>+5.57</b> | 20.7 |
| 0.5     | .59   | 5.71                    | <b>+3.46</b> | 11.3 |
| 1.0     | .70   | 3.67                    | <b>+2.05</b> | 5.4  |
| 1.5     | .77   | 2.56                    | +0.58        | 1.1  |
| 2.0     | .82   | 1.62                    | −0.03        | −0.1 |
| 3.0     | .88   | 0.02                    | −2.28        | −7.9 |

port lowers the loss on race Other at every skew and under both stepsizes: .073 against .119 for group-blind averaging on the prevalence instance (full coverage .031), and .031 against .035 on the aligned one, where it matches full coverage. Amer-Indian, the other single-group race, behaves the same way. On IMDb-Wiki the top-importance identities show the crossover of Table 2 one step earlier: under decay transport wins at  $\nu \leq .70$  (77.0 against 86.4 at  $\nu=.31$ , full 72.2; 73.6 against 79.9 when aligned) and loses at the three most infeasible skews, while at constant stepsize it wins only at  $\nu=.31$ . The overall objective and the rare-group loss therefore tell the same story, with the rare group the more sensitive of the two.

**E1: the headline cells.** Table 1 reports the four instances a practitioner would recognize. Transport wins on three: Adult under prevalence-proportional availability, 0.2269 against 0.2479 ( $t=45$ ), and both IMDb cells, 86.06 against 91.63 ( $t=21$ ) and 84.28 against 86.37 ( $t=13$ ). On Adult with importance-aligned availability the two coincide at the oracle value, which is what Proposition 2 predicts when  $\tilde{p} = p$ : there is nothing to correct and transport does not manufacture a difference. The fixed multiplier diverges on both aligned instances and is far off on the other two, so naive rescaling is no alternative to solving (4).

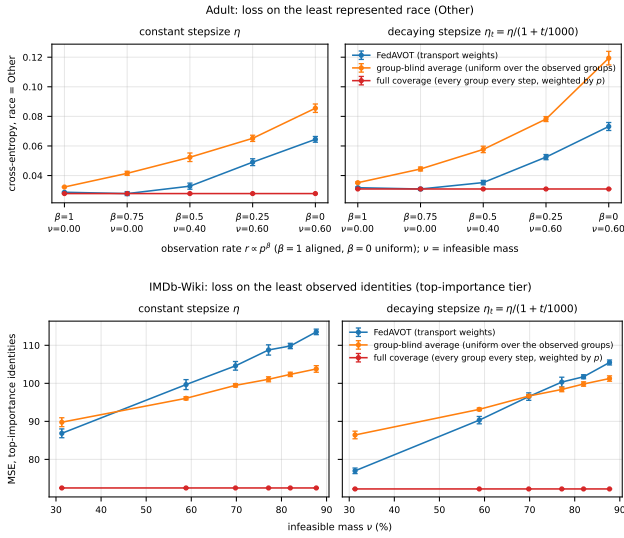


Figure 2: Loss on the least represented group, the metric the fairness target is about. Top, Adult: cross-entropy on race Other, one group of 30 out of 100, to which the category-uniform target assigns  $p_i = 0.2$ ; the axis is the observation-rate exponent,  $\beta=1$  aligned,  $\beta=0$  uniform (prevalence). Bottom, IMDB-Wiki: MSE on the 20 highest-importance identities, the least observed ones once  $\beta > 0$ , against  $\nu$ . Left panels constant stepsize, right panels decaying. Same protocol as the tables ( $K=3$ ,  $H=5$ , 4000 steps, tail-500 mean  $\pm$  std over 5 seeds); lower is better. Curves are the group-wise entries of the same runs as Table 1 and Fig. 1.

**E2: the crossover, and where the sign comes from.** Table 2 and Fig. 1 are the paper’s main evidence. On Adult, whose five skews are omitted from the table, the measured gain tracks  $\Delta_{\text{floor}}$  within 4% at three of five (.0134 against .0129, .0050 against .0049, .0002 against .0003) and exceeds it at the two most infeasible (.0210 against .0110, .0181 against .0144), group-blind averaging sitting further above its floor. On IMDB-Wiki the gain is uniformly below the floor by an excess optimization residual that is close to constant in  $\beta$ , mean 1.9 in MSE units, and the measured sign flips precisely where  $\Delta_{\text{floor}}$  crosses that value, between  $\beta=1.5$  (2.56, gain +0.58) and  $\beta=2.0$  (1.62, gain −0.03). This is Theorem 8 read at finite  $T$ : the sign of the comparison is a quantity one computes in advance.

**E3: the  $\ell_1$  bound is too loose to be the criterion.** The distances  $\|p - \hat{p}\|_1$  and  $\|p - \tilde{p}\|_1$  are also computed on every instance, and  $\|p - \hat{p}\|_1 < \|p - \tilde{p}\|_1$  holds in all eleven, including the three where transport loses or ties. The  $\ell_1$  form of Theorem 8 therefore never predicts a loss and cannot serve as the criterion; the exact floors can.

**E4: the residual is a stepsize artifact.** The left panel of Fig. 1 runs the identical instances at constant stepsize. There transport loses on all six IMDB skews, by as much as 7.9, even at  $\nu=.31$  where the floors are 6.99 apart. The transport weights  $Y[\cdot, j]$  concentrate mass on rarely observed groups, so although Lemma 6 bounds the second moment identically for both rules, the variance about the mean is larger, and at constant stepsize the iterates sit on a noise floor proportional

to it. Under  $\eta_t = \eta/(1 + t/1000)$  that floor decays, the measured curves fall onto their dashed floors, and every sign reversal attributable to the stepsize disappears. The residual is a property of the schedule, not of the estimator.

**E5: the decomposition in closed form.** The mirrored IMDB-Wiki instance ( $\nu=.88$ ) is where Theorem 8 can be read off exactly. The Sinkhorn limit truncates the target at the observation-rate ceiling,  $\hat{p}_i \approx \min(p_i, r_i)$  up to renormalization (maximum deviation .023), with  $\|p - \hat{p}\|_1 = 1.58$  of a possible 2. The frontier is least squares, so  $\Phi(p) = 82.9$ , matching the measured full plateau 82.95, and  $\Phi(\hat{p}) = 105.9$ : the alignment floor is 23.0 and the realized bias equals its exact expression  $|(p - \hat{p})^\top L(\theta^\dagger)| = 30.7$ , where  $L$  collects the per-group risks, while the  $\ell_1$  form  $2M\|p - \hat{p}\|_1$  is conservative since  $M \geq \max_i L_i(\theta^\dagger) \approx 454$ . Group-blind averaging has the same floor to two decimals,  $\Phi(\tilde{p}) = 105.9$ : both surrogates starve the same groups, so  $\Delta_{\text{floor}}$  vanishes and the entire measured gap between the two rules at this endpoint of Table 2 is residual, which is why it is the one instance where transport still loses by a significant margin under decay ( $\beta=2$ , where the floors are 1.6 apart, is a tie).

**E6: risk aversion, and where it does not help.** Sweeping  $(\alpha, \gamma)$  on a  $10 \times 10$  grid, the overall objective is minimized on the  $\gamma = 1$  edge on every instance, which is  $\kappa(1, \gamma) = 1$  made visible and confirms that risk aversion costs the average. The one place the CVaR layer earns its keep is the worst race on Adult under prevalence availability, where GAVOT+CVaR at  $(0.2, 0.7)$  reaches 0.3624 against 0.3679 for GAVOT and 0.3994 for group-blind averaging. Per race, transport improves four of five (Black .179 vs .200, Asian-Pac .368 vs .399, Am-Ind .175 vs .189, Other .073 vs .119) and regresses only the majority group (.339 vs .332), which is what reweighting toward a category-uniform target is supposed to do.